Indian Institute of Technology Roorkee

Department of Applied Mathematics and Scientific Computing

Evaluating CNN and ViT on Skin Cancer Detection

A Summer Internship Report

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Certificate

This is to certify that the internship report entitled “Evaluating CNN and ViT on Skin Cancer Detection”, submitted by Khushi Jha, is a record of the work carried out by her during the Summer Internship Programme (25 May 2026 – 17 July 2026) at the Department of Applied Mathematics and Scientific Computing, Indian Institute of Technology Roorkee, under my supervision.

The work presented in this report, to the best of my knowledge, is an authentic account of the intern’s own study, implementation, and analysis, and has not been submitted elsewhere for the award of any other degree or diploma.

Prof. Millie Pant

Supervisor

Place: Roorkee Date: July 17, 2026

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Acknowledgement

Seven weeks ago, “Vision Transformer” was just a term I had seen in a paper title and quietly hoped nobody would ask me to explain. Writing that sentence now, at the end of this report, is a good way to measure how much this internship actually gave me.

I am deeply grateful to my supervisor, Prof. Millie Pant, for giving me the room to learn a fairly large field from first principles instead of jumping straight to results, and for the kind of feedback that pushes you to think rather than just to fix a plot.

I would like to sincerely thank my mentor, Mr. Shubham Joshi, for the many patient conversations about attention mechanisms, broken training runs, and the occasional confusion between a tensor’s shape and my own expectations of it. His guidance shaped not just what I built, but how I approached debugging, reading papers, and being honest about what did and did not work.

I also thank the Department of Applied Mathematics and Scientific Computing, IIT Roorkee, for hosting this internship and providing the computational resources without which none of the experiments in this report would exist.

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Khushi Jha

Abstract

Skin cancer, and melanoma in particular, is one of the few cancers where the difference between an early diagnosis and a late one can be the difference between a minor procedure and a life-threatening illness. Yet the standard diagnostic tools available to a clinician – heuristics such as the ABCDE and CASH rules – still leave unaided visual inspection accuracy in the range of 60–80%. This gap is precisely where computer vision has spent the last decade

trying to help, first with Convolutional Neural Networks (CNNs), and more recently with Vision Transformers (ViTs).

This report documents a seven-week summer internship focused on understand- ing, implementing, and critically evaluating this progression – from classical image processing to CNNs, to ViTs, to a hybrid CNN-Transformer multi-modal archi- tecture. The internship began with the fundamentals of computer vision (edge detection, convolution, pooling), moved through transfer learning and the Vision Transformer architecture, and then applied both DenseNet121 and ViT-B/16 as baseline classifiers on the MILK10k (ISIC 2025) dataset. In parallel, a recent multi-modal explainable-AI framework from the literature, XAAI-Ledger, was re- produced from its published methodology on the ISIC 2019 dataset, including its Grad-CAM and SHAP based explainability components.

The results tell a fairly honest and, at times, humbling story: reproduced accuracy fell well short of the numbers reported in the source paper, largely due to class imbalance and the limited compute and time available within an internship timeline – a finding that is, in itself, a useful and common lesson in applied deep learning. The report closes by proposing a concept-bottleneck extension for future work, aimed at making the model's diagnostic reasoning traceable to clinically meaningful concepts rather than an opaque probability score.

This document is organised as a standard internship report, with Chapter 1 introducing the problem and motivation, Chapter 2 covering the theoretical back- ground required to understand every architecture used later, and the remaining chapters (methodology, experiments, results, and future work) building on this foundation.

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CHAPTER 1

Introduction

Every year, a huge number of skin lesions get looked at, squinted at, photographed, and sometimes biopsied – all to answer one deceptively simple question: is this dangerous, or not? Most of the time the answer is no. But the small fraction of times it is yes, and specifically when that "yes" is melanoma, timing matters more than almost anything else in oncology.

This chapter sets up why that tim- ing problem exists, why artificial intelligence has become a serious candidate for closing the gap, and what this internship specifically set out to do about it.

1.1 Artificial Intelligence in Healthcare

AI in healthcare did not begin with deep learning, even though it can feel that way from the inside of 2026. The earliest serious attempt at bringing “intelligence” into clinical decision-making was MYCIN in the 1970s – a rule-based expert system that reasoned about bacterial infections using a set of hand-coded if-then rules. It worked reasonably well on paper, and essentially never left the paper, because rule systems are brittle: every new edge case needs a new rule, and medicine is mostly edge cases.

The next real shift came with statistical machine learning in the 1990s and 2000s, particularly in radiology, where Computer-Aided Diagnosis (CAD) systems started flagging suspicious regions in mammograms and chest X-rays using hand- engineered features fed into classifiers like support vector machines. These systems were an improvement, but they were only ever as good as the features a human decided to extract – and deciding which features matter for a disease that presents in a thousand different ways is, itself, an extremely hard problem.

Deep learning changed the question entirely. Instead of an expert manually de- ciding that “asymmetry” or “edge irregularity” matters, a convolutional network could be shown thousands of labelled images and left to discover which patterns ac- tually correlate with the diagnosis. The 2012 ImageNet moment (AlexNet) is usu- ally cited as the spark for the broader field, and by 2016 the same idea had reached dermatology directly, with CNN-based classifiers matching board-certified derma- tologists on curated benchmarks. Vision Transformers entered medical imaging a few years later, bringing a different inductive bias – global self-attention instead of local convolution – and the last few years have pushed toward multi-modal,

explainable systems that try to combine image evidence with patient metadata and justify their own predictions. Figure 1.1 lays this progression out on a single timeline. URL 🔗

Figure 1.1: A brief evolution of AI in healthcare, from rule-based expert systems to modern multi-modal, explainable clinical AI.

It is worth being honest about what CAD systems actually are before we go further, because the term gets thrown around loosely. A CAD system does not make a diagnosis. It makes a suggestion – a second opinion that a clinician is free to override. This distinction matters a lot for how the rest of this report should be read: none of the models discussed here are being proposed as replacements for a dermatologist. They are being evaluated as candidates for exactly this second- opinion role.

1.2 Computer Vision in Medical Imaging

Medical image analysis has walked the same path that computer vision walked more broadly, just a little later and a little more cautiously (understandably – mistakes here have real consequences). That path looks roughly like Figure 1.2: it starts with traditional, hand-crafted image processing, moves into general deep learning, narrows into convolutional architectures built specifically for images, and most recently widens again into transformer-based architectures that were originally designed for language. URL 🔗

Figure 1.2: The broad evolution of computer vision techniques used in medical imaging, from hand-crafted filters to attention-based transformer architectures.

Traditional computer vision relied on filters designed by people – edge detec- tors, blob detectors, texture descriptors – which is precisely the kind of hand- engineering that made early CAD systems brittle. Deep learning replaced hand- designed features with learned ones, CNNs specialised that idea for grid-structured

data like images, and Vision Transformers are the most recent attempt at asking whether a more general, attention-based architecture can do the same job while modelling relationships across the entire image rather than a local neighbourhood. Chapter 2 is dedicated entirely to unpacking every stage of this diagram in techni- cal detail; this section exists only to give the reader a map before we start walking through it. URL 🔗

1.3 Skin Cancer Detection

Skin cancers broadly split into two families. Non-melanoma skin cancers – primarily basal cell carcinoma (BCC) and squamous cell carcinoma (SCC) – are common, usually slow-growing, and rarely fatal if treated. Melanoma is far rarer but disproportionately dangerous: it accounts for a small fraction of skin cancer cases but the large majority of skin-cancer deaths, because it can metastasize to other organs if left untreated.

This is exactly why early diagnosis is the single biggest lever available. Lo- calised melanoma, caught before it spreads, has a five-year survival rate above 99%. Once it has spread to distant organs, that number falls below 30%. There is no equivalent jump anywhere else in the treatment pipeline – no drug, no surgical technique – that moves survival rates this much. The entire value proposition of an AI screening tool for skin cancer rests on this one fact: if it can nudge diagnosis earlier, even slightly, it is doing something that really matters.

Clinically, dermatologists use two well-known heuristics to structure their vi- sual inspection, both referenced directly on the ISIC dataset documentation and in the wider dermatology literature:

- ⊠ ABCDE rule – Asymmetry, Border irregularity, Color variation, Diam- eter (typically > 6 mm), and Evolution (change over time) [2]. URL 🔗
- ⊠ CASH rule – Color, Architecture, Symmetry, and Homogeneity, often used as a complementary framework in dermatoscopic assessment [3]. URL 🔗

Figure 1.3 sketches the ABCDE criteria schematically – not as clinical pho- tographs, but as a simplified visual mnemonic, which is really all the rule is meant to be in the first place. URL 🔗

The catch, and the reason this whole report exists, is that these rules depend entirely on the skill and consistency of the person applying them. Two differ- ent clinicians looking at the same lesion do not always agree, and inter-observer variability is a well-documented problem in dermatology, with even experienced clinicians' unaided accuracy sitting around 60–80% [1]. This is the gap AI is being URL 🔗

Figure 1.3: Schematic illustration of the ABCDE rule used in manual dermato- logical screening (Asymmetry, Border irregularity, Color variation, Diameter, Evo- lution). Illustration is a simplified mnemonic diagram, not a clinical image.

asked to fill: not to replace clinical judgement, but to offer a second, consistent opinion that does not get tired, does not have an off day, and has effectively seen more lesions than any single clinician ever will.

1.4 Problem Statement

Framed plainly: manual visual screening for melanoma is inconsistent, heuristic- based, and caps out at around 60–80% accuracy even among experienced clinicians [1]. Deep learning models promise a more consistent alternative, but in practice they inherit a set of problems that are easy to state and surprisingly hard to fix: URL 🔗

- ⊠ Visual similarity across classes. Early melanoma and a benign nevus can look almost identical to the naked eye (and, it turns out, to a CNN too) – the differences are often subtle textural cues rather than obvious shape differences.
- ⊠ Class imbalance. Malignant cases are, thankfully, rare in the population – which is great for public health and genuinely unhelpful for training a classifier. Figure 1.4 shows exactly how lopsided this gets on the ISIC 2019 dataset used later in this report: benign cases outnumber melanoma cases by more than four to one. URL 🔗

- ☒ Limited and inconsistent data. Dermoscopic datasets vary in imaging device, lighting, patient demographics, and label quality, which makes a model trained on one dataset fragile when it meets another.
- ☒ Metadata ignorance. A large share of published skin-cancer classifiers are image-only, even though a dermatologist would never ignore a patient’ s age, sex, or the anatomical site of a lesion when forming a judgement.
- ☒ Lack of explainability. A model that outputs “87% melanoma” with no justification is not something a clinician can act on with confidence – and regulatory approval for clinical AI increasingly requires some form of interpretability.

Figure 1.4: Class imbalance in the ISIC 2019 dataset used in this internship: 20,809 non-melanoma samples versus 4,522 melanoma samples – a ratio of roughly 4.6:1.

Every one of these five problems reappears later in this report, either as a design consideration in the models that were built, or as a direct explanation for why the reproduced results in later chapters fall short of the numbers reported in the literature.

1.5 Motivation

It would be easy to say “CNNs work well for images, so let’ s use a CNN,” and honestly, that was more or less my starting assumption before this internship. A few weeks in, that assumption did not survive contact with the literature or the data.

CNNs are excellent at picking up local texture – edges, colour gradients, small- scale patterns – because that is exactly what a convolutional filter is built to do. But melanoma diagnosis often depends on relationships across the whole lesion: is the colour distribution consistent from one edge to the other, does the border look different on one side than the other. A convolution kernel, by design, only looks at a small neighbourhood at a time; it has to stack many layers before it can reason about the image as a whole, and even then, that global reasoning is implicit and diluted.

Vision Transformers were the obvious next thing to try, because self-attention lets every patch of the image attend to every other patch from the very first layer – global context, right from the start. But ViTs are notoriously data-hungry, and the datasets available in dermatology, while not tiny, are nowhere close to the scale ViTs were originally designed for. So neither architecture, taken alone, is obviously “enough.”

The piece that convinced me the field is moving in the right direction, though, was metadata. A 45-year-old patient with a changing lesion on sun-exposed skin is a different clinical picture from a 20-year-old with a stable lesion on a rarely- exposed site – even if the two lesions look nearly identical in a photograph. Image- only models are, by construction, blind to this information. That is the real mo- tivation behind this internship: not just to benchmark CNN against ViT, but to understand why the field is increasingly moving toward architectures that fuse visual features with structured clinical metadata, and to reproduce one such ar- chitecture end-to-end to see, honestly, how well it holds up outside its original paper.

1.6 Objectives of the Internship

Rather than framing this as a research contribution with novel objectives, it is more honest to frame it the way it actually happened – as a structured learning path with concrete checkpoints:

- 1. Build a solid working understanding of classical and deep-learning-based computer vision, starting from edge detection up to convolutional networks.
- 2. Understand why CNNs work the way they do – not just how to call Conv2D in a framework, but what the convolution and pooling operations are actually doing to an image.
- 3. Study the Vision Transformer architecture in enough depth to explain patch embedding, positional encoding, and multi-head self-attention without reach- ing for a slide deck.
- 4. Implement two independent baseline classifiers – DenseNet121 and ViT-B/16 – on a modern multi-class dermatology dataset (MILK10k), and compare them fairly under identical training conditions.
- 5. Reproduce a recent published multi-modal, explainable architecture (XAAI- Ledger) from its methodology section alone, including its Grad-CAM and SHAP explainability components.
- 6. Explore what a meaningful next step beyond this architecture could look like, grounded in the specific weaknesses observed during reproduction.

1.7 Internship Timeline

The seven weeks of the internship were structured to build up complexity gradu- ally – theory before implementation, single-modality baselines before multi-modal fusion, and reproduction before proposing anything new. Table 1.1 and Figure 1.5 summarise this plan. URL 🔗

Table 1.1: Week-by-week internship work plan.

Week Work Done
Week 1 Computer Vision Basics
Week 2 Vision Transformer
Week 3 DenseNet Baseline
Week 4 ViT Baseline
Week 5 XAAI-Ledger Implementation
Week 6 Proposed Improvements
Week 7 Analysis

Internship Timeline 7-Week Work Plan

Figure 1.5: Seven-week internship timeline, from foundational computer vision concepts through to final analysis.

The chapters that follow roughly mirror this timeline. Chapter 2 covers ev- erything learned in Weeks 1–2; the baseline implementations, the XAAI-Ledger reproduction, and the proposed improvements and analysis from the later weeks are covered in the chapters that follow it. URL 🔗

CHAPTER 2

Background Study

This chapter is, essentially, seven weeks of notes cleaned up into something read- able. It has nothing to do with skin cancer specifically – that connection is made later. What it covers is the computer vision toolbox itself: how images were processed before deep learning existed, why convolutional networks became the default choice once deep learning arrived, and how Vision Transformers arrived to challenge that default. By the end of this chapter, every architecture used later in this report should feel like a natural extension of something explained here, rather than a black box pulled off a shelf.

2.1 Traditional Computer Vision

Before a neural network ever learns a single feature on its own, it is worth under- standing what a feature even is, because traditional computer vision made this completely explicit. An image, at its core, is just a grid of numbers – pixel inten- sities. Traditional CV techniques are, almost without exception, small mathemat- ical operations applied to this grid to pull out something structurally meaningful: edges, corners, blobs, textures.

2.1.1 Edge Detection

An edge is a place in the image where intensity changes sharply – the boundary of an object, say, or the border of a lesion. Mathematically, an edge corresponds to a large gradient in the image intensity function I(x, y). Three filters come up again and again:

Sobel operator. Approximates the image gradient in the x and y directions using two small 3×3 kernels:

The overall edge strength (gradient magnitude) at each pixel is then

|G| = q G2 x +G2 y.

Gaussian blur. Before edge detection, images are usually smoothed with a Gaussian kernel to suppress noise, since raw gradients are extremely sensitive to it:

Laplacian operator. Rather than a first-order gradient, the Laplacian is a second-order derivative operator, useful for detecting regions of rapid intensity change in all directions at once:

Figure 2.1 shows all three of these operators applied to the same test image, computed directly for this report rather than borrowed from a paper, so the reader can see exactly what each filter does to the same input. URL 🔗

Figure 2.1: Gaussian smoothing, Sobel edge magnitude, and Laplacian filtering applied to the same grayscale test image, computed for this report.

The honest limitation of this entire family of methods is that a human had to decide, in advance, which filter matters. Sobel is good at edges. It is not good at detecting a subtle texture change that does not manifest as a sharp gradient. This is exactly the ceiling that deep learning eventually broke through – by learning the filters instead of hand-designing them.

2.2 Deep Learning Basics

An Artificial Neural Network (ANN) is, structurally, a sequence of layers, where each layer computes a weighted sum of its inputs followed by a non-linear activation function:

a(l) = σ

where W(l) and b(l) are the learnable weights and bias of layer l, and σ is a non- linearity such as ReLU. Stack enough of these layers and, in principle, the network can approximate extremely complex functions – this is the universal approximation

W(l)a(l−1) + b(l)⧄ ,

intuition behind deep learning.

The problem with applying a plain ANN directly to images is scale. A modestly sized 224 × 224 RGB image has 224 × 224 × 3 = 150,528 input values. A fully- connected first layer with even 1,000 neurons would need over 150 million weights just for that one layer – computationally wasteful, and worse, it throws away the fact that nearby pixels are related to each other. A Convolutional Neural Network (CNN) fixes both problems at once, by replacing full connectivity with small, shared filters that slide across the image (the same idea as the Sobel filter above, except now the filter' s numbers are learned rather than hand-picked). This gives CNNs two properties that make them well suited to images specifically:

- ⧄ Local connectivity – each neuron only looks at a small patch of the image, which matches the intuition that a pixel is most related to its neighbours.
- ⧄ Weight sharing – the same filter is reused across the entire image, drasti- cally reducing the number of parameters and letting the network detect the same pattern (e.g. an edge) regardless of where it appears.

2.3 Convolution Operation

The convolution operation itself is simple enough to compute by hand for a small example, which is usually the fastest way to actually understand it. For a single- channel input I and a kernel K of size k × k, the output at position (i, j) is

Three extra parameters control how this operation behaves in practice, and all three matter a great deal for the architectures used later in this report:

- ⧄ Kernel size – the spatial extent of the filter (commonly 3 × 3 or 5 × 5). Larger kernels see more context per step but cost more computation.
- ⧄ Stride – how far the kernel moves between applications. A stride of 2 roughly halves the output' s spatial dimensions.

- ☒ Padding – whether the input is padded with zeros at the border, so the kernel can be centred on edge pixels too ("same" padding keeps the output size equal to the input size; "valid" padding shrinks it).

For an input of size W×W, kernel size F, stride S, and padding P, the output

spatial size is

Figure 2.2 illustrates a 3 × 3 kernel sliding over a padded input with stride 1, producing one output feature map. [URL 🔗]

Figure 2.2: A 3×3 kernel sliding with stride 1 over a zero-padded input, producing a single output feature map. The highlighted receptive field on the input maps to a single highlighted output pixel.

Pooling. Convolutional layers are almost always followed by a pooling oper- ation, which reduces the spatial resolution of the feature map while keeping the most important information. Max pooling, the most common choice, simply takes the maximum value within each small window:

Pooling gives the network a small amount of translation invariance (a feature is still detected even if it shifts slightly) and reduces the parameter count of subsequent layers.

A typical CNN, then, is simply a repeated stack of Convolution → Activation → Pooling blocks, with the spatial size shrinking and the number of channels (feature types) growing at each stage, before a final classification head flattens everything into class probabilities.

2.4 Transfer Learning

Training a CNN or a ViT completely from scratch requires a very large labelled dataset – ImageNet, the standard benchmark, contains over a million labelled images across a thousand categories. Dermatology datasets, by comparison, are small. Transfer learning sidesteps this problem with a simple observation: the early layers of a network trained on natural images learn very general features (edges, colours, simple textures) that are useful for almost any visual task, not just the one the network was originally trained on.

The typical recipe is:

- 1. Pretraining – train a large CNN or ViT backbone on a large, general dataset (ImageNet).
- 2. Feature extraction – freeze the pretrained backbone's weights and use it purely to convert an image into a feature vector.

- 3. Fine-tuning – optionally, unfreeze some or all of the backbone and continue training on the target dataset (here, dermoscopic images) at a low learning rate, so the general features adapt slightly to the new domain without being destroyed.

Figure 2.3: The transfer-learning recipe used throughout this report: a frozen, ImageNet-pretrained backbone (CNN or ViT) acts as a general-purpose feature ex- tractor, feeding into a small trainable head adapted to the skin-cancer classification task.

Every baseline model built during this internship – DenseNet121, ViT-B/16, and the EfficientNetB0 backbone used inside the XAAI-Ledger reproduction – follows exactly this pattern, and this is the reason it is possible to get reasonable performance on a dataset with only tens of thousands of images rather than a million.

2.5 Vision Transformer

The Vision Transformer (ViT) takes the transformer architecture, originally built for language, and applies it to images with a surprisingly small number of image- specific changes [4]. The core idea: treat an image as a sequence of patches, exactly the way a sentence is a sequence of words, and let self-attention handle the rest. URL 🔗

2.5.1 Patch and Position Embedding

An input image of size H × W × C is split into N non-overlapping patches of size P × P, where N = HW P2 . Each flattened patch is linearly projected into a D-dimensional embedding:

where E is the learnable patch-embedding matrix, xclass is a special learnable CLS token prepended to the sequence (its final-layer representation is later used for classification), and Epos is a learnable positional embedding – necessary because, unlike a CNN, self-attention has no inherent notion of spatial order and needs to be told where each patch came from.

2.5.2 Transformer Encoder

The embedded sequence passes through L identical encoder blocks, each built from two sub-layers wrapped in residual connections and Layer Normalization:

where MSA is Multi-Head Self-Attention and MLP is a small feed-forward network (typically two linear layers with a GELU activation in between). The residual connections are what make it possible to stack many such blocks without gradients vanishing, and LayerNorm keeps the activations at each layer well-scaled during training.

2.5.3 Multi-Head Self-Attention

Self-attention is the mechanism that lets every patch look at every other patch. For a single attention head, each input token is projected into a Query, Key, and Value vector, and attention weights are computed as a scaled dot-product:

where dk is the dimensionality of the key vectors (the scaling factor prevents the dot products from growing too large and saturating the softmax). Multi-head attention simply runs h of these attention operations in parallel with different learned projections, and concatenates the results – allowing the model to attend to different types of relationships (e.g. colour similarity in one head, spatial proximity in another) simultaneously.

The upshot of all this machinery is that, from the very first encoder block, the CLS token can attend to every patch in the image directly. This is a fundamentally different inductive bias from a CNN, where a neuron in an early layer can only see a small local neighbourhood and has to wait for many layers of convolution before its receptive field covers the whole image.

Figure 2.4: The Vision Transformer (ViT) pipeline: an image is split into patches, linearly embedded with a prepended CLS token and positional embeddings, passed through L transformer encoder blocks (each with multi-head self-attention, residual connections, and an MLP sub-layer), and the final CLS token representation is classified by a small MLP head.

2.6 CNN vs ViT: A Theoretical Comparison

With both architectures now unpacked, it is worth putting them side by side before any experiment is run. Table 2.1 summarises the theoretical trade-offs; the empirical side of this comparison, on an actual dermatology dataset, is picked up again later in this report. URL 🔗

Neither column in Table 2.1 is strictly better than the other – and that, ulti- mately, is the whole justification for the multi-modal, hybrid approach explored in the rest of this report. A CNN backbone is well suited to extracting fine local tex- ture from a dermoscopic image; a transformer-based fusion layer is well suited to relating that visual evidence to structured clinical metadata as a global reasoning step on top. Chapters that follow build directly on both halves of this table – first evaluating DenseNet121 (a pure CNN) against ViT-B/16 (a pure transformer) as independent baselines, and then examining an architecture that tries to combine the strengths of both. URL 🔗

Table 2.1: Theoretical comparison between Convolutional Neural Networks and Vision Transformers.

Aspect	CNN	ViT

Inductive bias	Strong: locality, weight shar-	Weak: minimal built-in as-
	ing, translation invariance	sumptions about image struc-
		ture
Receptive field Local at early layers; grows		Global from the very first
	gradually with depth	layer, via self-attention
Data require-	Performs well on small-to-	Typically needs large-scale
ment	medium datasets	pretraining data to match
		CNN performance
Parameter effi-	Fewer parameters for a given	Generally more parameters;
ciency	receptive field, due to weight	self-attention is quadratic in
	sharing	sequence length
Interpretability	Grad-CAM, saliency maps	Attention-map visualisation
tools	(well established)	(more direct, since attention
		weights are inherently inter-
		pretable)
Typical strength Fine, local texture and edge		Long-range dependencies and
	patterns	global context

Computational	Convolution is linear in image	Self-attention scales quadrati-
cost	size	cally with number of patches

CHAPTER 3

Literature Review

Chapter 2 built the toolbox – convolution, transfer learning, self-attention. This chapter puts that toolbox in context by asking a narrower question: what has actually been tried, by other people, for this specific problem, and what did they find? The chapter moves from general CNN-based skin cancer work through Vision Transformers, into explainability and multimodal fusion, and closes with a detailed review of the one paper this internship set out to reproduce – XAAI-Ledger – before stating, explicitly, the gap that motivates the implementation chapters that follow.

3.1 Deep Learning in Skin Cancer Detection

The shift from hand-crafted features to learned ones, introduced in general terms in Section 2.1, reached dermatology specifically once large public datasets became available. Romero-Lopez et al. [2] were among the early groups to show that a CNN trained directly on dermoscopic images could outperform classical feature-plus-SVM pipelines on lesion classification, without any hand-engineered ABCDE-style features at all. This result mattered less for its raw accuracy number, which has since been surpassed many times over, and more for what it demonstrated in principle: that a network could discover diagnostically relevant patterns – asymmetry, border irregularity, colour variation – on its own, directly from pixels, essentially rediscovering the ABCDE heuristic from data rather than being told it.

What followed this initial demonstration was less a single breakthrough than a fairly steady widening of scope – larger and more diverse datasets, deeper architectures, and, eventually, an entirely different way of processing an image altogether (Vision Transformers, discussed in Section 3.3). The rest of this chapter follows that widening in roughly the order it happened.

3.2 CNN-Based Skin Cancer Classification

Once CNNs were established as viable, the natural next question was which CNN. DenseNet [5], used later in this report as Baseline 1, connects every layer to every subsequent layer in a feed-forward fashion, which encourages feature reuse and, in practice, tends to need fewer parameters than an equally deep architecture without such connections for a comparable level of accuracy. This property is particularly attractive in dermatology, where labelled data is comparatively scarce and a smaller, well-regularised network is less prone to overfitting than a very large one.

A separate line of work focused on combining textural and colour-based features with an ensemble of CNN backbones rather than relying on a single network. Sharafudeen and Chandra [3] proposed a CASH-rule-based architecture that fuses texture features with a deep backbone specifically to keep the model's reasoning aligned with a criterion a dermatologist would already recognise, and a related ensemble-based follow-up [11] combined multiple CNN backbones with textural descriptors for improved robustness on abnormality detection. The consistent finding across this body of work is that CNN accuracy on curated, single-source benchmarks (such as HAM10000) is high – frequently exceeding 85–90% – but drops noticeably when the same model is evaluated on a dataset collected with a different imaging device or patient population, a generalisation gap that reappears directly in the results of this report (Chapter 5).

Table 3.1 summarises the CNN-based approaches discussed above.

Table 3.1: Comparison of representative CNN-based approaches to skin lesion classification.

Study	Backbone	Dataset	Reported strength	Reported limitation
Romero-Lopez et al. [2]	Custom CNN	Dermoscopic subset	First demonstration of learned-feature superiority over hand-crafted features	Small dataset, single-source
Huang et al. [5] (DenseNet)	DenseNet-121/169/201	ImageNet + transfer	Parameter-efficient, strong feature reuse via dense connectivity	Not dermatology-specific; needs fine-tuning
Sharafudeen and Chandra [3, 11]	CASH-rule + texture fusion	Curated dermoscopic set	Keeps model reasoning aligned with a clinical heuristic	Ensemble adds inference cost and complexity

3.3 Vision Transformer-Based Classification

Once ViTs [4] demonstrated competitive image classification performance without any convolutional inductive bias, dermatology researchers began asking whether the same held for skin lesions specifically. The general finding, echoed across several follow-up studies, is a variant of the trade-off already summarised theoretically in Table 2.1: a ViT trained from scratch on a dermatology-sized dataset underperforms a CNN of comparable size, but a ViT that starts from an ImageNet-pretrained checkpoint and is fine-tuned – exactly the transfer-learning recipe of Section 2.4 – closes most or all of that gap, and in several published comparisons slightly exceeds the CNN baseline on global, texture-independent cues such as overall lesion shape and colour distribution.

Hybrid CNN-transformer encoders, first popularised outside dermatology by architectures such as TransUNet [9] for medical image segmentation, extended this idea further by using a CNN as a feature extractor and a transformer as the layer that reasons globally over those features, rather than choosing one architecture family exclusively. This "CNN-for-locality, transformer-for-globality" pattern recurs throughout the multimodal literature reviewed in Section 3.5, and is the direct conceptual ancestor of the fusion architecture reproduced in Chapter 6.

3.4 Explainable AI in Medical Imaging

A classifier that only outputs a probability is difficult to act on clinically, a point already raised in Section 1.4. Two explainability techniques dominate the dermatology literature and are used directly in this report's own reproduction work (Chapter 6): Grad-CAM [8] and SHAP.

Grad-CAM produces a coarse localisation heatmap by using the gradients flowing into the final convolutional layer to weight that layer's feature maps, highlighting the image regions that most influenced a given prediction. It is well suited to CNN-style architectures precisely because it needs a spatial convolutional feature map to operate on, and it has become close to a default requirement in published dermatology CNN papers as a sanity check that the network is attending to the lesion rather than to an imaging artefact (a ruler, a hair, a border marking pen).

SHAP takes a complementary, feature-level approach based on Shapley values from cooperative game theory: it estimates each input feature's marginal contribution to the final prediction. For image-only models SHAP is less commonly used than Grad-CAM, but for the tabular metadata branch of a multimodal model – age, sex, anatomical site – it is close to the standard tool, since Grad-CAM has no equivalent for a non-spatial input. Table 3.2 summarises the two techniques side by side.

Table 3.2: Grad-CAM versus SHAP as explainability tools in this report.

Aspect	Grad-CAM	SHAP
Input type it explains	Spatial (image)	Tabular / structured
Mechanism	Gradient-weighted class activation	Shapley value attribution
Output	Pixel-level heatmap	Per-feature contribution score
Used on	CNN image branch	Metadata (age, sex, site) branch

Neither technique explains why the model considers a concept diagnostically relevant in the first place – both operate after the fact, on a model that was never asked to represent clinical concepts explicitly. This limitation is picked up directly in Section 3.7 and again in the future-work discussion of Chapter 7.

3.5 Multimodal Learning for Skin Cancer

The metadata-ignorance problem raised in Section 1.4 is precisely what multimodal architectures are designed to address. The general pattern across published multimodal skin-cancer work is a two-stream design: an image branch (typically a pretrained CNN or ViT) produces a visual embedding, a separate branch processes structured metadata (age, sex, anatomical site, sometimes patient history), and a fusion mechanism – concatenation, an MLP, or, increasingly, cross-attention – combines the two before a final classification head.

A separate and, in recent years, empirically dominant family of multimodal approaches avoids end-to-end fusion entirely and instead stacks a CNN's out-of-fold prediction probabilities with a gradient-boosted decision tree (GBDT) trained on tabular metadata. This two-stage "CNN + GBDT" pattern traces back to the winning solution of the SIIM-ISIC 2020 melanoma classification challenge (Ha, Liu, and Liu) and was the dominant pattern again among top entries to the ISIC 2024 "Skin Cancer Detection with 3D-TBP" challenge, which typically ensembled multiple CNN/ViT backbones with several GBDT variants (LightGBM, XGBoost, CatBoost) across many cross-validation folds. Baseline 3 in this report (Section 5.4) is a deliberately scaled-down single-backbone, single-GBDT reproduction of exactly this recipe, and its results are read directly against this literature in that section.

Across both fusion styles, the consistent published finding is that metadata helps – but only when it is genuinely complementary to the image, and only when there is enough of it. A handful of coarse fields (age, sex, one anatomical-site category) tend to add a small, sometimes negligible improvement over an image-only baseline; richer metadata (patient history, multiple lesion sites over time, dermatologist-recorded features) adds substantially more. This distinction turns out to matter directly for the results in this report, where the metadata available is limited to exactly the coarse end of that spectrum.

3.6 Review of XAAI-Ledger

The specific paper this internship set out to reproduce, referred to throughout this report as XAAI-Ledger, sits squarely within the end-to-end fusion family described above. Its architecture, discussed in full technical detail in Section 6.2, couples a frozen pretrained CNN image branch with a small MLP metadata branch, projects both into a shared representation, and fuses them through a multi-head self-attention block before a final sigmoid classification head for binary melanoma versus non-melanoma prediction. The paper reports Grad-CAM and SHAP explainability outputs as a core part of its contribution, framing the work as much around transparency as around raw accuracy.

The paper's headline results, reported on the ISIC 2019 dataset, are notably strong – accuracy and sensitivity figures in the high nineties, discussed in full in Table 5.7 and Table 6.2 alongside this internship's own reproduction of the same pipeline. What the published methodology section does not fully specify, and what turns out to matter a great deal in practice, is the exact class-imbalance handling strategy used during training; ISIC 2019 has a roughly 4.6:1 imbalance between non-melanoma and melanoma cases (Figure 1.4), and how that imbalance is compensated for – class weighting, oversampling, a custom loss – has a large effect on sensitivity specifically. Chapter 6 returns to this gap directly, since it is the leading candidate explanation for the difference between the paper's reported numbers and the numbers reproduced in this internship.

3.7 Research Gap

Pulling the threads of this chapter together, three gaps stand out as directly relevant to the work that follows:

- **Benchmark comparability.** Published CNN and ViT results for skin lesion classification are rarely evaluated under an identical protocol – same dataset, same split, same number of epochs, same evaluation metrics – which makes head-to-head comparisons across papers unreliable. Chapters 4 and 5 of this report address this directly by training DenseNet121 and ViT-B/16 under exactly matched conditions on the same dataset.
- **Reproducibility of multimodal fusion claims.** Multimodal architectures, including XAAI-Ledger, are frequently proposed and evaluated only by their original authors, with implementation details – imputation strategy, class-imbalance handling, exact fusion hyperparameters – often under-specified in the published methodology. Chapter 6

reproduces XAAI-Ledger independently from its methodology section and reports the reproduced numbers honestly, imbalance-related shortfall included.

- **Interpretability that stops at attribution.** Grad-CAM and SHAP, reviewed in Section 3.4, explain a trained model's existing decision after the fact; neither forces the model to reason through clinically meaningful concepts in the first place. This is the specific gap the Concept Bottleneck extension proposed in Chapter 7 is aimed at.

These three gaps motivate the structure of the rest of this report directly: Chapters 4 and 5 build and fairly compare the two single-modality baselines, Chapter 6 reproduces and honestly evaluates the multimodal reference architecture, and Chapter 7 proposes a concrete next step grounded in the third gap above.

CHAPTER 4

Dataset and Experimental Setup

Chapter 3 closed by identifying a comparability gap between published CNN and ViT results. This chapter describes exactly how that gap was closed within this internship – the datasets used, how they were prepared, and the precise experimental conditions under which every model in Chapter 5 was trained – so that the results reported there are fully reproducible from this description alone.

4.1 Experimental Workflow

All experiments in this report follow the same four-stage pipeline: dataset acquisition and cleaning, preprocessing, model training under a fixed protocol, and evaluation against a common set of metrics. Figure 4.1 lays this pipeline out end to end; the sections that follow expand each stage.

[Insert Figure:
Description: End-to-end experimental workflow used for both baseline models – dataset loading, cleaning/imputation, image preprocessing and augmentation, model training loop, and evaluation.
Suggested image: Custom workflow diagram (create using a flowchart tool such as draw.io,

based on the four stages described in Section 4.1) Source: Original diagram, not sourced online]

4.2 Dataset Description

Three public dermatology datasets were used across this internship, each serving a distinct role. Table 4.1 summarises all three, alongside ISIC 2020, which was reviewed during dataset selection but not used for training.

Table 4.1: Datasets used or evaluated during this internship.

Dataset (Year)	Images	Classes	Metadata	Class balance	Role in this report
ISIC 2019	25,331	8 (MEL, NV, BCC, AK, BKL, DF, VASC, SCC)	Age, sex, anatomic site	17.9% melanoma (4,522 / 25,331)	Training/validation set for the XAAI-Ledger reproduction (Chapter 6)
HAM10000	10,015	7 (akiec, bcc, bkl, df, mel, nv, vasc)	Age, sex, lesion localisation	11.1% melanoma (1,113 / 10,015)	Cited directly in XAAI-Ledger; reviewed for cross-dataset context
MILK10k (ISIC 2025)	10,480 (5,240 paired lesions)	11 broad categories (up to 48 fine-grained ISIC-DX diagnoses)	Age, sex, anatomic site, skin tone (6-level), histopathology flag	95.7% histopathology-confirmed ground truth	Training/validation set for DenseNet121 and ViT-B/16 baselines (Chapter 5)

MILK10k was selected as the primary baseline dataset because it is the most recent public release with genuine dual-image (clinical close-up plus dermoscopic) multimodality and explicit skin-tone annotation, making it the natural target for the concept-bottleneck extension proposed in Chapter 7. ISIC 2019 was retained specifically for the XAAI-Ledger reproduction, since it is the dataset used in the original paper.

4.3 Dataset Analysis

MILK10k's 11 diagnostic categories are not evenly represented; several rare classes contain fewer than 20 samples in the training split, which reappears as a direct explanation for near-zero per-class F1 scores in Section 5.4. ISIC 2019 has a coarser, 8-class label set and the roughly 4.6:1 non-melanoma-to-melanoma imbalance already illustrated in Figure 1.4.

[Insert Figure:
Description: Bar chart of per-class sample counts for MILK10k (11 diagnostic categories), highlighting the classes with fewer than 20 training samples.
Suggested image: Custom bar chart generated from the MILK10k class-distribution data used during this internship
Source: Original chart, not sourced online
]

4.4 Data Preprocessing

Both datasets went through a common preprocessing pipeline before being passed to a model:

- **Image resizing.** All dermoscopic images were resized to $224 \times 224 \times 3$ to match the input resolution expected by the ImageNet-pretrained DenseNet121, ViT-B/16, and EfficientNetB0 backbones used throughout this report.
- **Normalisation.** Pixel values were rescaled and normalised using the same per-channel mean and standard deviation each backbone was originally pretrained with, so that the pretrained low-level filters (Section 2.4) remain well matched to the input distribution.
- **Metadata imputation.** Missing tabular fields were filled using median imputation for numeric fields (age) and mode imputation for categorical fields (sex, anatomical site) – the same strategy used in the XAAI-Ledger reproduction (Section 6.3), applied here for consistency across the report.
- **Metadata encoding.** Categorical metadata fields were one-hot encoded, and continuous fields were scaled with a MinMaxScaler, prior to being concatenated or fed into any

metadata-consuming branch (Baseline 3, Section 5.4).

- **Train/validation split.** A stratified split was used throughout, preserving class proportions in both partitions – 75/25 for DenseNet121 and ViT-B/16 on MILK10k, and 80/20 for Baseline 3 on MILK10k (Section 5.4).

Table 4.2 summarises the resulting split sizes.

Table 4.2: Train/validation split sizes used across baseline models.

Model	Dataset	Split ratio	Validation set size
DenseNet121	MILK10k	75/25	2,620
ViT-B/16	MILK10k	75/25	2,620
Baseline 3 (CNN + GBDT)	MILK10k	80/20	2,096
XAAI-Ledger reproduction	ISIC 2019	75/25 (stratified)	6,333 (5,202 non-melanoma, 1,131 melanoma)

4.5 Experimental Environment

All models were trained using the computational resources provided by the Department of Applied Mathematics and Scientific Computing, IIT Roorkee, using standard deep-learning frameworks with GPU acceleration.

Table 4.3: Experimental environment.

Component	Configuration
Framework	PyTorch / TensorFlow (ImageNet-pretrained backbones)
Compute	GPU-accelerated training environment
Image backbones	DenseNet121, ViT-B/16, EfficientNetB0 (all ImageNet-pretrained)
Metadata model	Dense MLP (64 → 128 → 256)
Tabular baseline	XGBoost

4.6 Evaluation Metrics

Every classifier in this report is evaluated on a common set of metrics, chosen specifically because accuracy alone is misleading on an imbalanced dataset (Section 1.4):

- **Accuracy** – overall proportion of correct predictions; reported for comparability with prior literature, but not treated as the primary metric given class imbalance.
- **Macro-F1** – the F1 score averaged uniformly across all classes rather than weighted by class frequency, which prevents a model from scoring well purely by favouring the majority class.
- **Macro/ROC-AUC** – the area under the ROC curve, averaged across classes, used as a threshold-independent measure of separability.
- **Sensitivity (recall) and specificity** – reported specifically for the binary melanoma/non-melanoma task in Chapter 6, since a missed melanoma (false negative) and a false alarm (false positive) carry very different real-world costs.

4.7 Hyperparameter Configuration

Table 4.4 lists the training hyperparameters used for the two primary baselines in Chapter 5, held identical across both models specifically so that the comparison in Section 5.4 isolates the effect of architecture rather than training configuration.

Table 4.4: Hyperparameter configuration for DenseNet121 and ViT-B/16 baselines.

Hyperparameter	Value
Backbone initialisation	ImageNet-pretrained
Epochs	20
Train/validation split	75/25
Input resolution	224 × 224 × 3
Metadata used	Age, sex, anatomical site
Dataset	MILK10k (ISIC 2025)

With the dataset, preprocessing, and training protocol now fixed, Chapter 5 applies this exact setup to build and compare the two baseline classifiers.

CHAPTER 5

Benchmark Model Implementation

Chapter 4 fixed the dataset and the experimental protocol. This chapter applies that protocol to build two independent single-modality baselines – DenseNet121 and ViT-B/16 – under identical training conditions on MILK10k, and reports a comparative analysis between them. This is the empirical counterpart to the theoretical CNN-versus-ViT comparison already laid out in Table 2.1.

5.1 Need for Benchmark Models

Before attempting any multimodal fusion architecture (Chapter 6) or interpretability layer (Chapter 7), it is necessary to establish how well the two underlying image architecture families perform on their own, under a shared, controlled protocol. Without this step, any later claim that fusion or metadata "helps" would have nothing fair to be measured against – a concern raised directly in Section 3.7 as a broader comparability gap in the published literature. DenseNet121 and ViT-B/16 were chosen specifically because they represent the two architecture families discussed throughout Chapter 2: a mature, parameter-efficient CNN, and a global-attention transformer, both initialised from ImageNet pretraining (Section 2.4).

5.2 DenseNet121 Baseline

Architecture overview. DenseNet121 [5] connects each layer to every other layer in a feed-forward fashion within a dense block, so that each layer receives the feature maps of all preceding layers as input. This encourages feature reuse and gives the network strong gradient flow during training, at a lower parameter cost than a comparably deep architecture without such connections – the theoretical property discussed generally in Section 2.3 and revisited here in its specific, published form.

[Insert Figure:
Description: DenseNet-121 architecture diagram showing dense blocks, transition layers, and the growth-rate connectivity pattern.
Suggested image: Original DenseNet architecture figure from Huang et al., "Densely Connected Convolutional Networks," CVPR 2017

Source: arXiv:1608.06993 / official CVPR paper figure
]

Implementation. An ImageNet-pretrained DenseNet121 backbone was fine-tuned as an image-only classifier on MILK10k, following the transfer-learning recipe of Section 2.4, with age, sex, and anatomical site available in the dataset but not consumed by this particular baseline (it is deliberately image-only, to isolate the CNN architecture's own performance before any metadata is introduced).

Training configuration. Table 5.1 lists the configuration used, matching Table 4.4.

Table 5.1: DenseNet121 training configuration.

Parameter	Value
Dataset	MILK10k (10,480 images, 11 classes)
Split	75/25 train-validation
Epochs	20
Initialisation	ImageNet-pretrained
Modality	Image-only

Results. Table 5.2 reports validation performance.

Table 5.2: DenseNet121 validation results on MILK10k.

Metric	Value
Accuracy	71.18%
Macro-F1	66.75%
Macro-AUC	0.9016

[Insert Figure:
Description: Confusion matrix for the DenseNet121

11-class MILK10k baseline (from the internship presentation).
Suggested image: Confusion matrix figure generated during this internship
Source: Internship presentation, Slide 5
]

Observations. The model performs strongest on visually distinctive classes and struggles most on classes with overlapping visual presentation – melanoma versus melanocytic nevus (MEL vs NV), and basal cell carcinoma versus actinic keratosis (BCC vs AK) – consistent with the visual-similarity problem identified generally in Section 1.4 and with the CNN limitations reviewed in Section 3.2.

5.3 Vision Transformer Baseline

Architecture overview. ViT-B/16 [4] applies the patch-embedding, positional-encoding, and multi-head self-attention pipeline described in full in Section 2.5, using 16×16 patches and the "Base" configuration (12 encoder blocks). Its defining property relative to DenseNet121 is global receptive field from the first layer, as summarised theoretically in Table 2.1.

[Insert Figure:
Description: ViT-B/16 architecture diagram (patch embedding, positional encoding, transformer encoder stack, classification head).
Suggested image: Original ViT architecture figure from Dosovitskiy et al., "An Image is Worth 16x16 Words," ICLR 2021
Source: arXiv:2010.11929
]

Implementation. ViT-B/16, initialised from ImageNet pretraining, was fine-tuned as an image-only classifier under the exact same protocol as DenseNet121 – same dataset, same split, same number of epochs – so that any performance difference between the two models can be attributed to architecture rather than training conditions.

Training configuration. Table 5.3 lists the configuration, identical to Table 5.1 except for the backbone itself.

Table 5.3: ViT-B/16 training configuration.

Parameter	Value
Dataset	MILK10k (10,480 images, 11 classes)
Split	75/25 train-validation (same split as DenseNet121)
Epochs	20
Initialisation	ImageNet-pretrained
Modality	Image-only

Results. Table 5.4 reports validation performance.

Table 5.4: ViT-B/16 validation results on MILK10k.

Metric	Value
Accuracy	75.50%
Macro-F1	72.31%
Macro-AUC	0.9251

Observations. ViT-B/16 outperforms DenseNet121 on every metric under this identical protocol. The gain is most visible on classes where the diagnostically relevant cue is spread across the lesion rather than concentrated in one local region – exactly the kind of global relationship the self-attention mechanism (Section 2.5.3) is built to capture, and exactly the limitation of local convolution raised as motivation in Section 1.5.

5.4 Comparative Analysis

Table 5.5 places both single-modality baselines side by side, and Table 5.6 extends the comparison to Baseline 3, a CNN + gradient-boosted decision tree (GBDT) stacking pipeline built to test whether adding metadata through a fusion-of-predictions approach (Section 3.5) improves on either image-only baseline.

Table 5.5: DenseNet121 vs ViT-B/16 on MILK10k (11-class, identical protocol).

Metric	DenseNet121	ViT-B/16
Accuracy	71.18%	75.50%
Macro-F1	66.75%	72.31%
Macro/ROC-AUC	0.9016	0.9251
Validation set size	2,620 (25% split)	2,620 (25% split)
Modality	Image-only	Image-only

Baseline 3 follows the ConvNeXt-Base out-of-fold (OOF) probability plus XGBoost stacking pattern reviewed in Section 3.5, using a single backbone and a single GBDT rather than the multi-backbone, multi-GBDT ensembles used by top entries in the ISIC 2024 challenge. Two pipeline variants were evaluated: Pipeline 1 uses only the CNN's OOF class probabilities plus metadata, and Pipeline 2 additionally engineers prediction-confidence features on top.

Table 5.6: Baseline 3 (CNN + GBDT stacking) results on MILK10k.

Metric	Pipeline 1 (OOF probs + metadata)	Pipeline 2 (+ confidence features)
Accuracy	69.27%	68.99% (−0.29 pt)
Macro-F1 (11-class)	0.39	0.38
Log-loss	not computed	0.971
Feature count	46	51

Both Baseline 3 pipelines trail the simple image-only baselines already established in Table 5.5. This is a direct, empirical instance of the metadata-richness caveat raised in Section 3.5: only three metadata fields (age, sex, anatomical site) are available, and this small a set carries little signal beyond what the CNN's own predicted probabilities already encode. Rare MILK10k classes with fewer than 20 training samples (Section 4.3) score close to zero F1 regardless of which pipeline is used, and a single backbone plus a single GBDT is simply not the same model as the multi-model ensembles used by the challenge winners this pattern is drawn from.

Table 5.7: Full ranking across all three benchmark models on MILK10k (11-class accuracy).

Rank	Model	Accuracy	Macro-F1	Modality
1	ViT-B/16	75.50%	72.31%	Image-only
2	DenseNet121	71.18%	66.75%	Image-only
3	Baseline 3 (CNN + GBDT)	69.27% (best pipeline)	0.39	Image + metadata (stacked)

The clearest finding of this chapter is a slightly uncomfortable one: on this dataset, under this protocol, adding metadata through prediction-level stacking did not help, and the simplest image-only ViT baseline remained the strongest model. This finding directly informs the design of the multimodal architecture reproduced in the next chapter – XAAI-Ledger fuses metadata at the representation level, through self-attention, rather than at the prediction level through stacking, which is a meaningfully different mechanism and is evaluated on its own terms there.

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